E. Tree

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Recently Bob invented a new game with a tree (we should remind you, that a tree is a connected graph without cycles): he deletes any (possibly, zero) amount of edges of the tree, and counts the product of sizes of the connected components left after the deletion. Your task is to find out the maximum number that Bob can get in his new game for a given tree.

Input

The first input line contains integer number n $(1 \le n \le 700)$ — amount of vertices in the tree. The following $n - 1$ lines contain the description of the edges. Each line contains the pair of vertices' indexes, joined by an edge, a_i, b_i $(1 \le a_i, b_i \le n)$. It's guaranteed that the graph described in the input is a tree.

Output

Output the only number — the maximum product of sizes of the connected components, that Bob can get after deleting some of the tree's edges.

Examples

input Copy

```
5
1 2
2 3
3 4
4 5
```

output Copy

```
6
```

input Copy

```
8
1 2
1 3
2 4
2 5
3 6
3 7
6 8
```

output Copy

```
18
```

input Copy

```
3
1 2
1 3
```

output Copy

```
3
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 23

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371973508	Apr/21/2026 15:51	Accepted
371973163	Apr/21/2026 15:48	Wrong answer on test 1
371973019	Apr/21/2026 15:46	Compilation error
371972838	Apr/21/2026 15:44	Compilation error

371972651	Apr/21/2026 15:42	Wrong answer on test 1

→ Problem tags

dp *2500

No tag edit access

→ Contest materials

- Announcement
- Tutorial

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Apr/21/2026 19:58:41 UTC+7 (n2).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by